

Compensation of the Refractive Index of Air in Laser Interferometer for Distance Measurement: A Review

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We review the progress made for compensation of the refractive index of air in laser-based distance measurements in the field of precision engineering. First, a comprehensive analysis is introduced to clarify how the overall measurement uncertainty is affected by the refractive index of the ambient air; particularly for dimensional metrology and geodetic survey using laser in open-air environment. Second, it is explained that the measurement uncertainty can be improved to a 10^{-8} level by adopting empirical dispersion formulae describing the relation of the laser wavelength with the environment parameters such as temperature, pressure, humidity and carbon dioxide concentration. Third, the principle of refractometers is given to describe that the measurement uncertainty can be enhanced to a 10^{-9} level by identifying the refractive index of air real time in well-controlled environmental conditions. Fourth, the two-color compensation method is discussed which enables precise laser-based distance measurement simply by using two different wavelengths without actual identification of the refractive index of open air. Finally, the recent approach of using femtosecond lasers is described with emphasis on the performance of the two-color method can be enhanced to a 10^{-9} level by stabilizing the pulse repetition rate as well as frequency stability of ultrashort pulse lasers with reference to the atomic clocks.

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NOMENCLATURE

L = measured distance
 e = excess fraction
 λ = laser wavelength
 n = refractive index of air
 T = air temperature
 P = air pressure
 P_w = water vapor pressure
 P_{CO_2} = CO₂ concentration

1. Introduction

In the field of precision engineering and metrology, laser interferometers are widely used for diverse applications as an important tool of distance and length measurement. Outstanding examples are axis positioning of precision machines and dimensional inspection of

machined parts.¹⁻³ Compared with other alternative sensors such as optical scales and electromagnetic transducers, laser interferometers are able to offer high precision with direct traceability to the physical definition of the metre using monochromatic lasers.⁴⁻⁶ Nonetheless, in order to realize extreme precision and accuracy particularly in the nanometer regime, special care is needed to carefully deal with the environmental disturbance that significantly influences the wavelength stability of laser sources.

Various interferometric techniques are being used in today's commercial laser interferometers, while the phase-measuring principle relying on homodyne⁷⁻⁹ or heterodyne¹⁰⁻¹³ laser sources is preferably adopted as it permits a high resolution less than a single wavelength of the light source. In this case, the measured distance L can be expressed in terms of the laser wavelength λ as $L = (m + e)\lambda$, with m being an integer multiple ($m = 0, 1, 2, \dots$) and e the excess fraction ($0 \leq e < 1$). Hence, for precise determination of L from the measured quantities of m and e , the true value of λ needs to be known very accurately, but it is not always the case in reality because the wavelength of air varies with many environmental parameters continuously in complicated ways.

For elaborate analysis of the effect of λ on the overall measurement

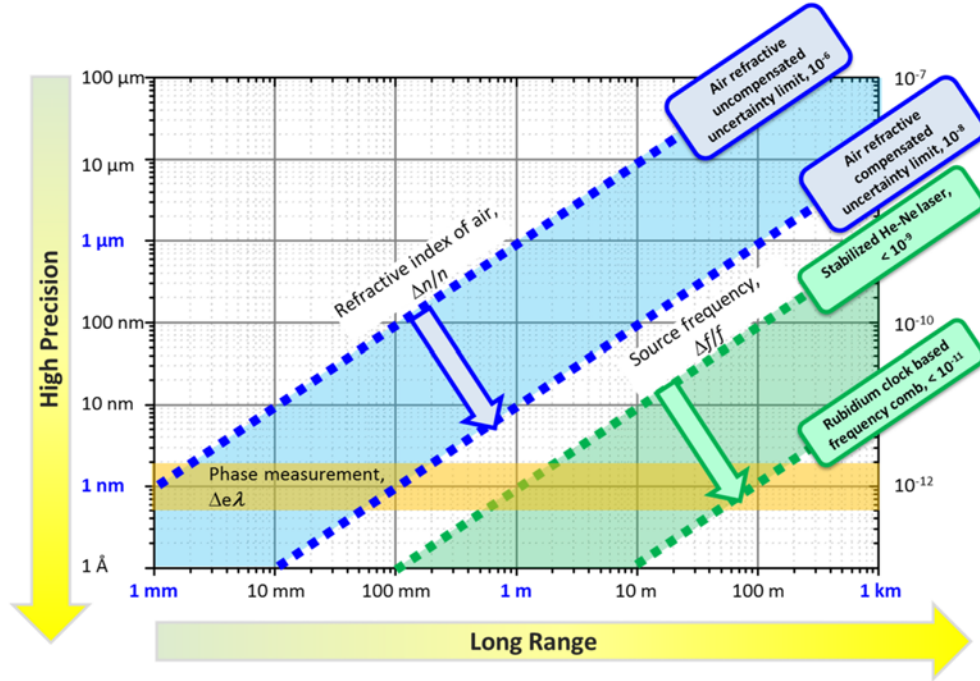


Fig. 1 Uncertainty diagram of laser interferometers dominated by three contributing parameters; the interferometric phase measurement error $\Delta e\lambda$, the fractional error in identifying the refractive index of air $(\Delta n/n)L$, and the fractional stability of the source frequency $(\Delta f/f)L$

uncertainty, the first-order perturbation of L with respect to λ can be taken as

$$\Delta L = (\Delta m + \Delta e)\lambda + (m + e)\Delta\lambda \approx \Delta e\lambda + (\Delta\lambda/\lambda)L \quad (1)$$

When high speed photo-detectors are used along with sufficiently fast electronics for data processing, the measured integer value of m may be assumed to have no error, i.e., $\Delta m = 0$. Then $\Delta e\lambda$ is related only to the subdivision capability of the interference fringe, which is usually much smaller than a single wavelength, i.e., $\Delta e \ll 1$. On the other hand, $\Delta\lambda$ represents the uncertainty in identifying the true value of λ , and the term $\Delta\lambda/\lambda$ denotes the fractional uncertainty of λ . Accordingly, the term $(\Delta\lambda/\lambda)L$ reflects the total error caused by $\Delta\lambda$, which is in proportion to the measured distance L . The laser wavelength is invariant in vacuum, being given as $\lambda_0 = c_0/f$; c_0 is the speed of light in vacuum and f is the optical frequency of the laser source. In air, depending on the ambient condition, the wavelength λ turns out to be no longer constant, deviating from the vacuum wavelength, i.e., $\lambda = \lambda_0/n = c_0/fn$ with n being referred to as the refractive index of air. The distance L is then expressed in terms of n and f explicitly as $L = (m+e)c_0/nf$. Thus, by extending Eq. (1), the uncertainty of L can be derived as

$$|u(L)|^2 = |\Delta e\lambda|^2 + \left| \left(\frac{\Delta n}{n} \right) L \right|^2 + \left| \left(\frac{\Delta f}{f} \right) L \right|^2 \quad (2)$$

Fig. 1 depicts how the overall uncertainty $u(L)$ is affected by three terms of $\Delta e\lambda$, $\Delta n/n$ and $\Delta f/f$ individually. First, $\Delta e\lambda$ remains small, independent of L , thus its contribution to $u(L)$ is meaningful only when the distance L is small. As L increases, $u(L)$ rises in proportion to L being dominated by either $\Delta n/n$ or $\Delta f/f$. The uncertainty level of $\Delta f/f$ in the state-of-the-art laser stabilization technology is typically in the range

of lower than 10^{-9} , which further reaches 10^{-11} or less when stabilized to the atomic clocks through the frequency comb of stabilized femtosecond lasers.¹⁴⁻¹⁶ On the other hand, the uncertainty of $\Delta n/n$ lies usually in the range of more or less 10^{-6} in uncontrolled environments, whereas it can be enhanced to 10^{-8} only when properly compensated. In consequence, $\Delta n/n$ turns out to be the most influencing parameter that has to be carefully treated to improve $u(L)$.

Fig. 2 depicts a chronological chart of the milestone achievements made during the last half century for the study of the refractive index of air since the first standard model was first established in 1953.¹⁷ Now it has been well established that $\Delta n/n$ can be calculated using empirical dispersion formula with concurrent sensing of the ambient parameters, thereby the measured distance can be corrected by post-processing.^{18,19} Or, the inaccuracy caused by the refractive index of air can be suppressed on site by installing a refractometer in parallel to the beam line of a laser interferometer.²⁰ These two approaches are known to provide the capability of identifying the refractive index of air with uncertainties near a $\sim 10^{-8}$ level in well-controlled homogeneous environmental conditions. However, in reality, the refractive index of air never be static due to inherent air turbulence, which make it difficult to identify the refractive index of air accurately all along the whole optical path to be measured. An alternative solution to cope with the problem is to employ two different wavelengths at the same time and eliminate the disturbance of the refractive index of air by making the most of the standardized dispersion formula of light. This two-wavelength or more preferably called two-color method is generally known to be effective for long-distance measurements in outdoor environments where air fluctuation is severe with non-homogenous and dynamical airflow.²¹⁻²³

In this review, we intend to describe how the refractive index of air has been dealt with so far in the field of optical distance metrology. In

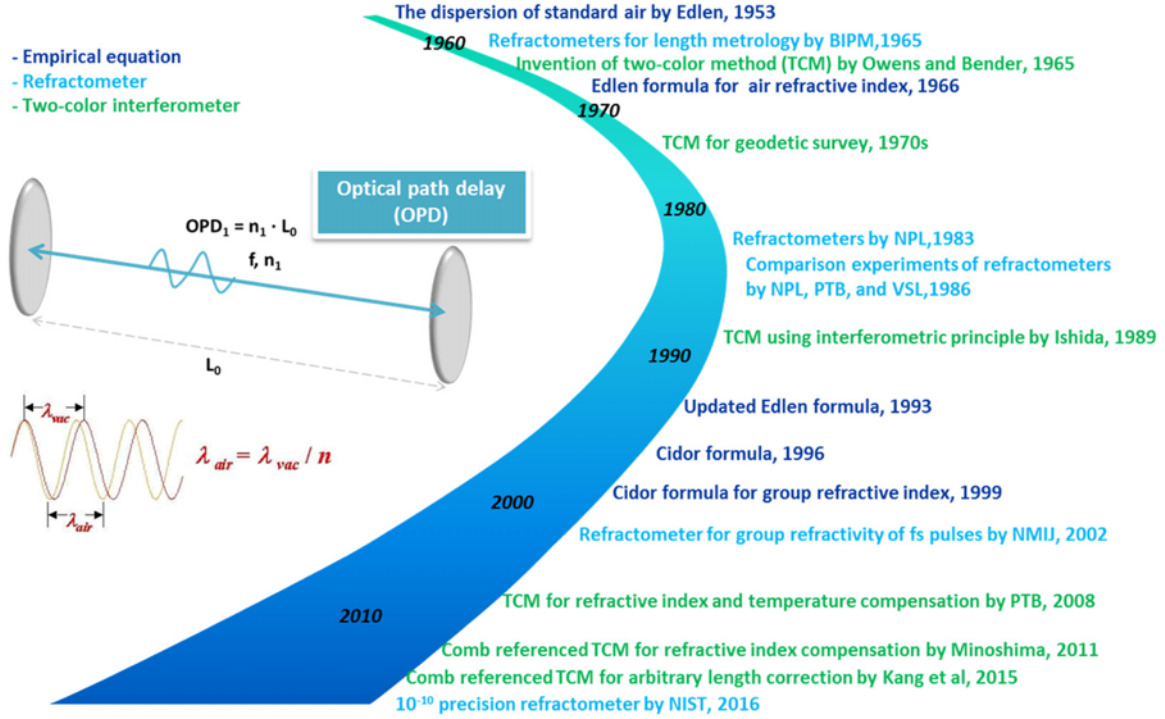


Fig. 2 Chronological summary of important events related to the progress made in chart for typical method of refractive index of air measurement; BIPM: Bureau international des poids et mesures, NPL: National Physical Laboratory, PTB: Physikalisch-Technische Bundesanstalt, VSL: Van Swinden Laboratorium, NMIJ: National Metrology Institute of Japan, NIST: National Institute of Standards and Technology

Section 2, two empirical equations are explained which have been established commonly for identification of the refractive index of air in terms of the ambient air parameters. In Section 3, the concept of refractometer is explained which has long been investigated for on-site determination of the refractive index of air. In Section 4, the two-color method is discussed which allows for self-correction of the refractive index of air real time during actual distance measurement. Finally, in Section 5, emphasis is put on describing the recent progress made by exploiting femtosecond lasers for accurate implementation of the two-color method, which is expected to pave the way towards the nanometer precision truly in optical distance measurement in the next generation precision engineering.

2. Empirical Formulae for the Refractive Index of Air

Being given as $\lambda = \lambda_0/n$, the light wavelength λ in air is always shorter than its vacuum wavelength λ_0 . This corresponds to the fact that light travels in air slower than in vacuum. Specifically, the refractive index n of air is larger than unity and its actual value varies with the wavelength λ as illustrated in Fig. 3(a). The λ -dependent of the refractive index is called dispersion, which causes a total deviation of $\sim 0.5 \times 10^{-6}$ when λ varies from 600-1600 nm. Dispersion appears to be steeper for small λ in the visible light range. In addition to dispersion, the refractive index is also affected by the environmental condition of the ambient air. Since the first empirical work on a standard model of air in 1953 was made,¹⁷ much analytical and experimental work has done to reach the general expression of the refractive index of air in the form of

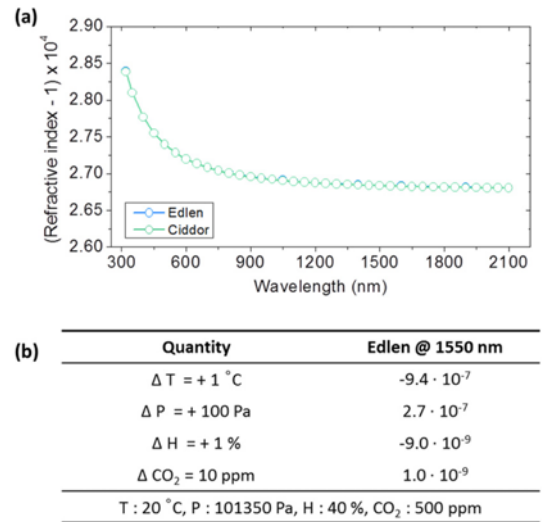


Fig. 3 (a) Variation of the refractive index of air with the wavelength, (b) Effects of four major influencing parameters of the ambient air on variation of the refractive index of air

$$n(\lambda, T, P, P_w, P_{CO_2}) - 1 = K(\lambda) \times D(T, P, P_{CO_2}, P_w) + P_w \times D_w(\lambda, T, P, P_{CO_2}, P_w) \quad (3)$$

The above equation implies that the refractive index is mostly influenced by four ambient parameters; temperature (T), pressure (P), CO₂ concentration (P_{CO_2}) and water vapor pressure (P_w). The effect of P_w is separated from other parameters and it may conveniently neglected

Table 1 Detailed expressions of the empirical formulae of the refractive index of air: Edlen vs. Ciddor equation

	Edlen formula ¹⁹⁻²²	Ciddor formula ²³⁻²⁵
$K(\lambda)$	$10^{-8} \times [8342.54 + 2406147 / \{130 - (\lambda/\mu\text{m})^{-2}\} + 15998 / \{38.9 - (\lambda/\mu\text{m})^{-2}\}]$	$10^{-8} \times [5792105 / \{238.0185 - (\lambda/\mu\text{m})^{-2}\} + 167917 / \{57.362 - (\lambda/\mu\text{m})^{-2}\}]$
$D(T, P, P_{\text{CO}_2}, P_w)$	$\{(P/\text{Pa}) / 96095.43\} \times \{[1 + 10^{-8} \times (0.601 - 0.00972 \times T/^\circ\text{C}) \times (P/\text{Pa})] / (1 + 0.0036610 \times T/^\circ\text{C})\} \times \{1 + 0.540 \times (P_{\text{CO}_2} - 0.0003)\}$	$\{[(1 - x_v) \times (P/\text{Pa})] / \{351.783223 \times Z_m(T + 273.15^\circ\text{C})/^\circ\text{C}\}\} \times \{1 + 5.34 \times 10^{-1} \times (P_{\text{CO}_2} - 0.00045)\}$
$D_w(\lambda, T, P, P_{\text{CO}_2}, P_w)$	$-10^{-10} \times \{292.75 / (T/^\circ\text{C} + 273.15)\} \times \{3.7345 - 0.0401 \times (\lambda/\mu\text{m})^{-2}\}$	$10^{-8} \times \{w_0 + w_1 \times (\lambda/\mu\text{m})^{-2} + w_2 \times (\lambda/\mu\text{m})^{-4} + w_3 \times (\lambda/\mu\text{m})^{-6}\} \times x_v \times (P/\text{Pa}) / [4.452450686 \times Z_m \times \{(T + 273.15^\circ\text{C})/^\circ\text{C}\}]$
$x_v = (P/P_w) \times [1.00062 + 3.14 \times 10^{-8} \times (P/\text{Pa}) + 5.60 \times 10^{-7} \times \{(T + 273.15^\circ\text{C})/^\circ\text{C}\}^2]$		
$a_0 = 1.58123 \times 10^{-6}, a_1 = -2.9331 \times 10^{-8}, a_2 = 1.1043 \times 10^{-10}, b_0 = 5.707 \times 10^{-6}, b_1 = -2.051 \times 10^{-8}, c_0 = 1.9898 \times 10^{-4}, c_1 = -2.376 \times 10^{-6}, d = 1.83 \times 10^{-11}, e = -7.65 \times 10^{-9}, Z_m = 1 - [(P/\text{Pa}) / \{(T/^\circ\text{C}) + 273.15\}] \times [a_0 + a_1(T/^\circ\text{C}) + a_2(T/^\circ\text{C})^2 + \{b_0 + b_1(T/^\circ\text{C})\}x_v + \{c_0 + c_1(T/^\circ\text{C})\}x_v^2] + [(P/\text{Pa}) / \{(T/^\circ\text{C}) + 273.15\}]^2 \times (d + e \times x_v^2)$		
$w_0 = 295.235, w_1 = 2.6422, w_2 = -0.03238, w_3 = 0.004028$		

for dry air. The sensitivity of each parameter to the refractive index at room temperature is given in Fig. 3(b), indicating that temperature is the most dominating parameter in practice.

Currently, for precise calculation of the refractive index of air using the ambient parameters, two empirical formulae are widely used as summarized in Table 1; one is the updated version of the Edlen equation²⁴⁻²⁶ as modified by Birch and Downs and the other is the Ciddor equation.^{19,27,28} No significant discrepancy exists between the two formulae, thus either one can be adopted for general uses. Nonetheless, one may expect the Edlen equation is suitable at room temperature near 20°C, while the Ciddor equation is more accurate under extreme environmental conditions usually for outdoor geodetic surveying.

Now an important issue is how accurately the empirical formulae are able to predict the true value of the refractive index in varying environmental conditions. It is generally accepted that the expanded uncertainty ($k = 2$) of both the equations is about 2×10^{-8} , but it deteriorates for extreme ambient conditions that are far from standard conditions. It is also worthwhile to note that major errors in predicting the refractive index using the empirical formulae come not from the equations themselves but from the inaccuracy in determining the ambient parameters. In other words, in order to exploit the empirical formulae in practice, it is essential to measure the ambient parameters as accurate as possible by selecting suitable sensors. Although the empirical equations are widely used for compensation of the refractive index of air, there are some practical limitations. Firstly, it requires a well-controlled homogeneous condition in the ambient air because sensing the ambient air parameter is usually confined to a point within the space where an interferometer is being used. The ambient air is usually not statically stable due to dynamic air turbulence, thus it is difficult to assume that the refractive index of air is precisely uniform along all optical paths. Secondly, it is practically not easy to realize real-time compensation since sensing the environment parameters takes a significant time particularly high precision is required.

3. Refractometers

The refractometer²⁹⁻³⁵ refers to interferometric devices that are configured to determine the refractive index of air directly on site in the environment where actual distance measurement using a laser interferometer takes place. As illustrated in Fig. 4, the refractometer is

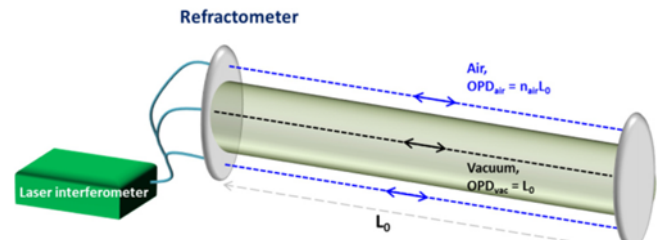


Fig. 4 Typical configuration of the refractometer

an interferometer itself that is designed to measure a fixed distance through a vacuum cell and at the same time through air using the same light source of the laser interferometer to be calibrated. Assuming the refractive index of the vacuum cell is exactly unity, the refractive index of air is determined by the ratio of the optical path length (OPL_{air}) measured in air to that of the vacuum cell (OPL_{vac}), i.e., $n = \text{OPL}_{\text{air}} / \text{OPL}_{\text{vac}}$. It is important to construct the vacuum cell very rigid, preferably in a symmetrical structure so that the mechanical deformation due to the pressure difference between vacuum and air can be minimal. Care is also needed to keep gravitational bending as small as possible so that the vacuum cell maintains the same length as the air path. In addition, thermal disturbance on the cell length needs to be minimized by using ultralow expansion (ULE) materials. The refractometer can also be used for experimental determination of the effect of a specific gas element on the refractive index. Quite a few refractometers based on the scheme of Fig. 4 are commercially available in the name of wavelength tracker. The Fabry-Perot (FP) cavity refractometer can be selected which provides a precise means of determining the refractive index of air by detecting the resonant frequency of the FP cavity using a tunable laser.³⁶⁻³⁹

It is worth noting that Fig. 5 shows an absolute type refractometer specially developed by Physikalische Technische Bundesanstalt (PTB) for precise determination of the contribution of water vapor to the refractive index of air.⁴⁰ This refractometer is equipped with additional devices to take control of the water vapor pressure within a temperature-controlled vacuum chamber. Adopting phase-stepping absolute interferometry, the refractometer is configured to monitor the length variation of a gage block of 299 mm length (Zerodur) as induced by the refractive index under control of the water vapor pressure. The light source includes 3 different wavelengths; 532 nm (Nd:YAG, I₂-stabilized),

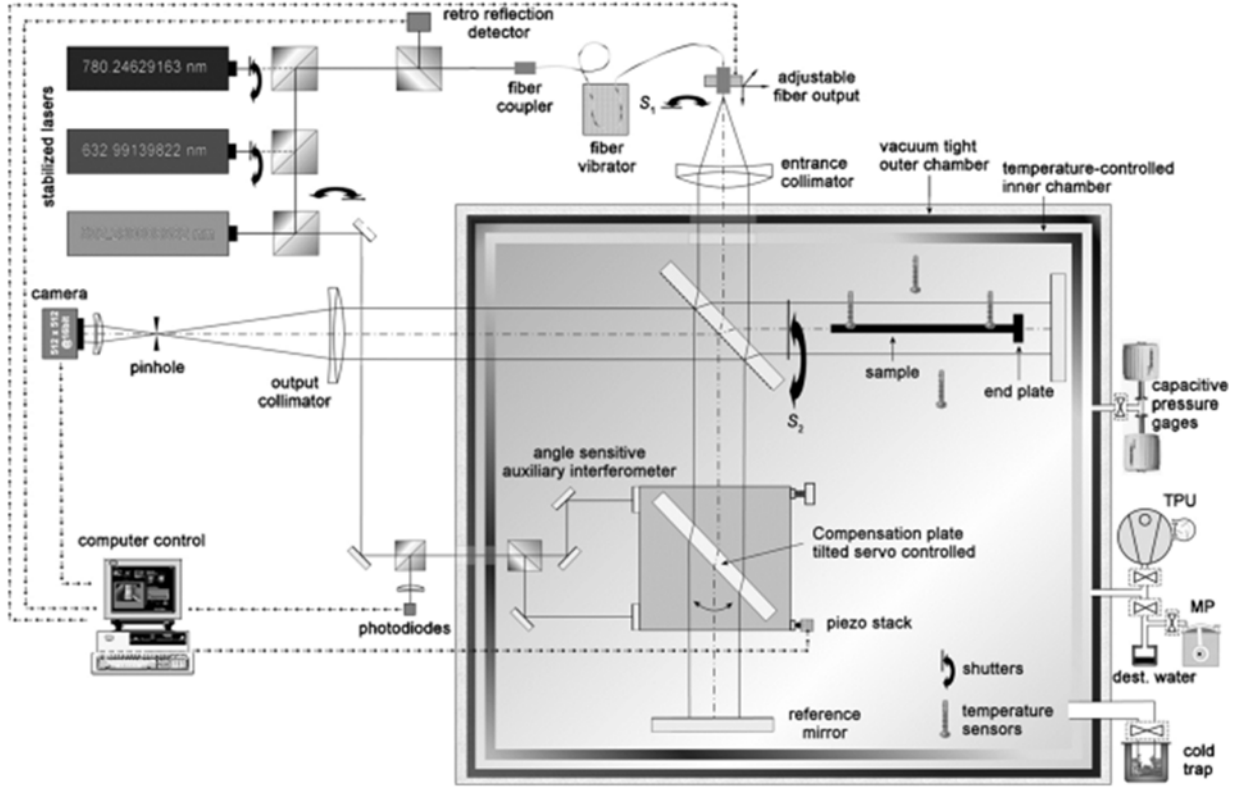


Fig. 5 Refractometer for determination of water vapor refractivity in PTB (Adapted from Ref. 40 with permission)

633 nm (He-Ne, I_2 -stabilized), and 780 nm (diode, Rb-stabilized) operating in a switching mode. The refractive index was identified as a function of water vapor with an uncertainty of $1.4\text{--}2.1 \times 10^{-9}$ in rms.

4. Two-Color Method

The two-color method (TCM) is intended for real-time correction of the refractive index of air in laser-based distance measurement. As depicted in Fig. 6, the two-color method is implemented by measuring a given distance with two different wavelengths of λ_1 and λ_2 simultaneously. Once the optical path lengths D_1 and D_2 are measured using λ_1 and λ_2 , the true geometrical length L_0 can be worked out as

$$L_0 = D_1 - A(D_2 - D_1), \text{ where } A = (n_1 - 1)/(n_2 - n_1) \quad (4)$$

It is important to note that n_1 and n_2 in Eq. (4) are the mean values averaged all along the optical path length for λ_1 and λ_2 . This implies that the effect of the refractive indices can be corrected even though they are not uniform within the optical path. The parameter A of Eq. (4) is comprised of n_1 and n_2 , so it is affected by the ambient temperature, pressure, humidity, and CO_2 concentration. In the dry air condition, however, the A parameter becomes dependent only on the vacuum wavelengths of λ_1 and λ_2 , since the influence of the ambient conditions is cancelled out as implied in Eq. (3). In reality, the atmospheric air contains some extent of water vapor and fails to meet the perfect dry air condition. Nonetheless, the effect of humidity is not significant compared with other ambient conditions, thus the A parameter may be

Two-color method

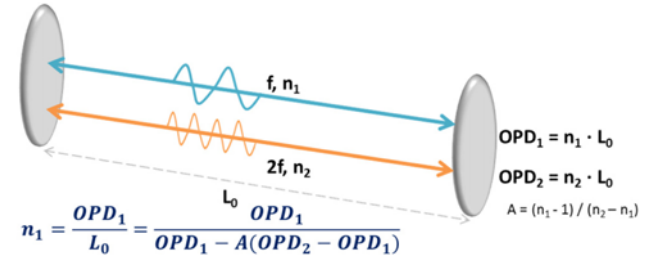


Fig. 6 Two-colour method for compensation of the refractive index of air in dry air

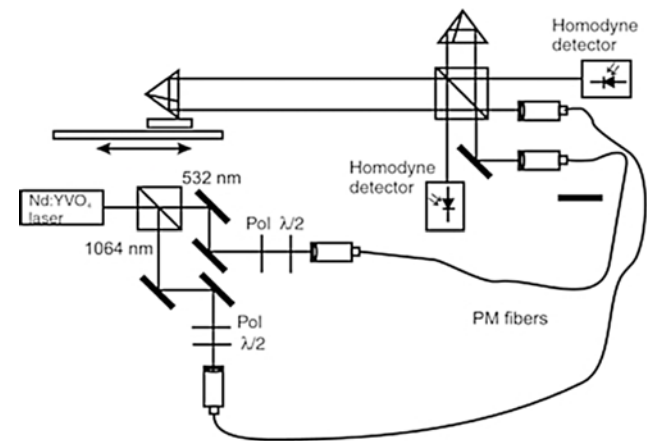


Fig. 7 Two-colour interferometer for distance measurement developed by PTB (Adapted from Ref. 55 with permission)

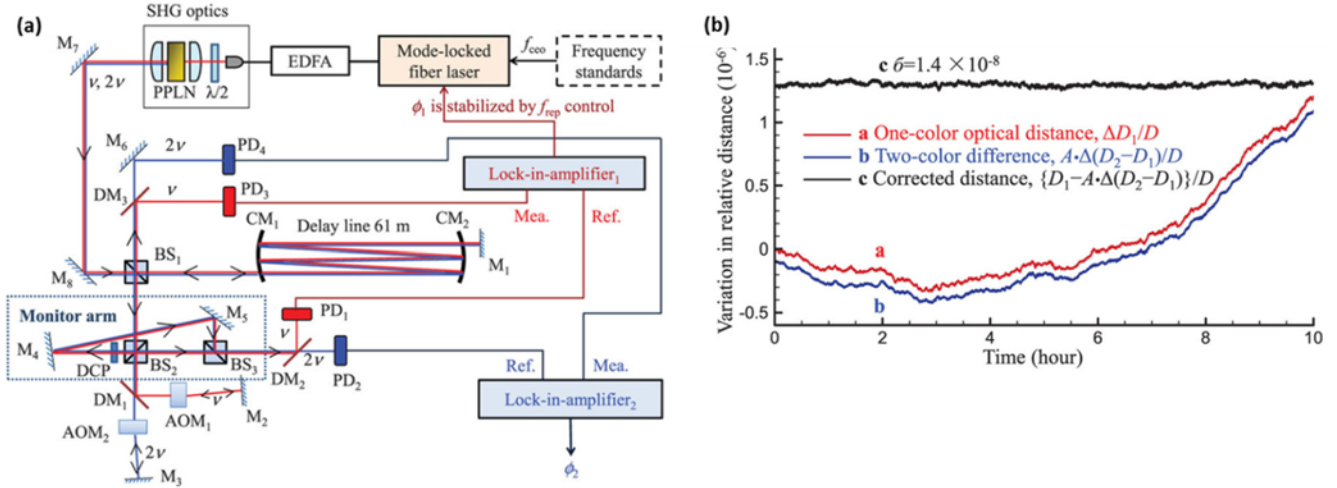


Fig. 8 Two-color pulse-to-pulse interferometer using a mode-locked femtosecond laser: (a) Experimental configuration, (b) Measurement result (Adapted from Ref. 67 on the basis OA)

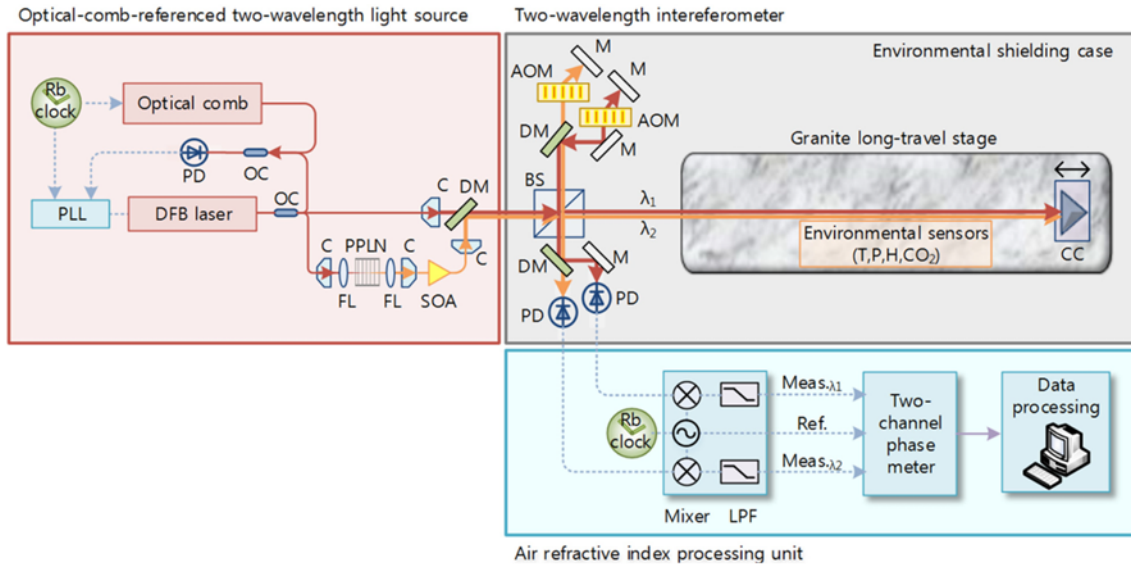


Fig. 9 Comb referenced two-wavelength interferometer developed in KAIST (Adapted from Ref. 73 on the basis OA)

assumed to be a constant being determined only by λ_1 and λ_2 .

The two-color method was first applied to compensate for the atmospheric fluctuation in geodetic survey with an improved uncertainty of 3.9×10^{-7} in measuring an open air distance of 5.9 km.⁴¹⁻⁴³ In geodetic survey, the light wavelength itself is too short to be used directly for long distance measurements, so various synthetic long wavelengths were generated by modulation of the polarization direction or intensity amplitude in the radio-frequency range. As a result, the measurement resolution was limited to a few millimeters.⁴⁴⁻⁴⁸ Meanwhile, the two-color method was first adopted for light wavelength interferometry in 1989,⁴⁹ and since then many attempts were made for real-time correction of the refractive index of air in high-precision laser-based distance measurements with ever growing laser technologies.⁵⁰⁻⁵⁴

Fig. 7 shows another scheme of two-wavelength interferometer that employs a Nd:YVO₄ laser (1064 nm) and its frequency-doubled laser (532 nm).⁵⁵ The A parameter turns out 65. The two wavelengths are

individually combined into a polarization-maintaining single-mode fiber. The target mirror is located at a 50 m distance. The ambient temperature was within 19.8 to 20.5°C with stability of $\pm 0.04^\circ\text{C}$. The relative humidity was regulated within 3-5% variation. The uncertainty over a 30 m distance was estimated to be $\pm 1.2 \times 10^{-7}$.

5. Frequency Comb for Self-Correction of the Refractive Index of Air

The frequency comb technology⁵⁶⁻⁵⁸ based on femtosecond lasers have brought about breakthroughs in diverse fields of metrology through precision frequency and time measurement.⁵⁹⁻⁶⁴ This is also true for two-color interferometry.⁶⁵⁻⁶⁷ As illustrated in Fig. 8(a), an Er-doped fiber laser is stabilized to the hydrogen maser so that the pulse repetition rate (f_r) is tunable near 54 MHz, which corresponds to a 5.5 m pulse-to-pulse

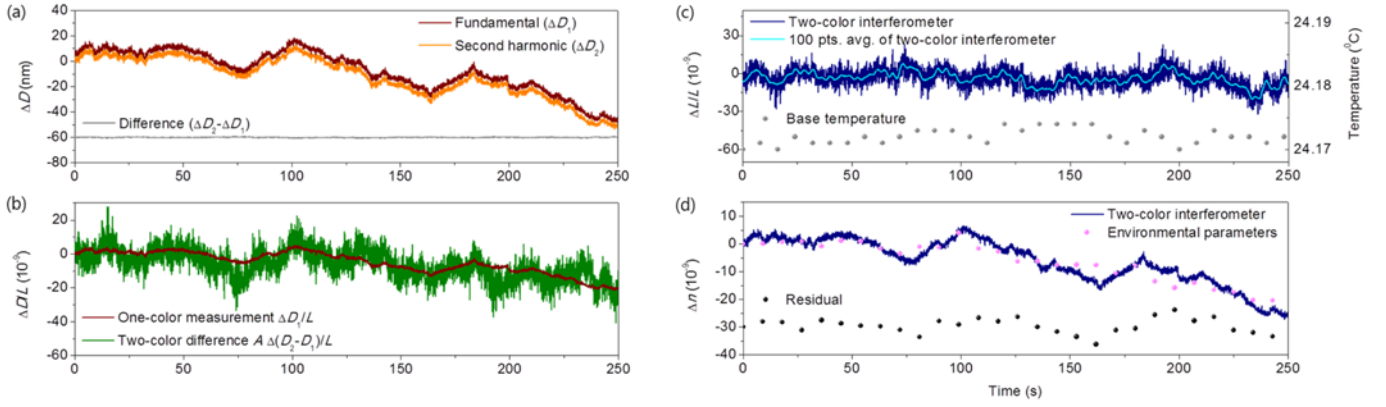


Fig. 10 Two-color measurement result for a given distance L of 2.5 m (Adapted from Ref. 73 on the basis OA)

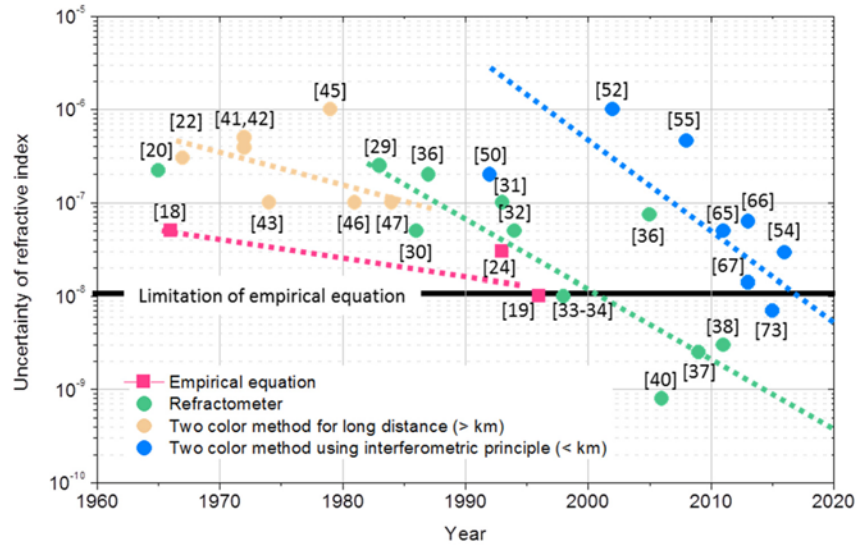


Fig. 11 State-of-the-art levels of uncertainty achieved for compensation of the refractive index of air

distance. The output beam is amplified by an Er-doped fiber amplifier (EDFA) and passes through a periodically poled LiNbO₃ (PPLN) crystal for the process of second harmonic generation (SHG). Then, the fundamental (ν) and second harmonic (2ν) beams are coaxially incident to an interferometer having a 61 m delay line. The ambient temperature is controlled within a thermally isolated box so that the geometric distance L is held nearly constant. The relative distance variation can be expressed as

$$\Delta L/L = \{\Delta D_1 - A\Delta(D_2 - D_1)\}/L, \text{ where } \Delta D_1 = -c\Delta f_r/f_r^2 \quad (5)$$

where c is the speed of light in vacuum. The variation of the pulse repetition rate (f_r) can be precisely measured with a frequency counter. Fig. 8(b) shows experimental results of self-corrected distance over 10 hours. The corrected distance ($\Delta L/L$) is shown to be nearly constant with a standard deviation of 1.4×10^{-8} while the refractive index of air varies about 1.4×10^{-6} over the same time scale. In this experiment, the measurement precision is estimated to be 10^{-8} , almost in the same uncertainty level of the empirical equations.

Fig. 9 shows the experimental configuration of the frequency comb

referenced two-wavelength interferometer. The measurement principle is based on phase measurement by heterodyne detection.^{68,69} The frequency comb of an Er-doped fiber femtosecond laser is stabilized to the Rb atomic clock.^{69,72} A fundamental wavelength (λ_1) at 1555 nm is generated using a distributed feedback (DFB) laser stabilized by the phase-locked locking (PLL) technique. A second wavelength (λ_2) is generated by frequency doubling of the fundamental wavelength so that $\lambda_2 = 2\lambda_1$ and amplified by a semiconductor optical amplifier (SOA). The two beams are simultaneously incident to the interferometer of Michelson type. The interferometer is contained inside a shield to minimize air turbulence and stabilize the ambient air temperature within ± 5 mK. Environmental sensors for on-side monitoring of temperature, air pressure, relative humidity, and CO₂ concentration are incorporated to determine the refractive index of air using the Ciddor's formula for comparison.⁷³

Fig. 10 presents experimental results for a given length of 2.5 m. The D_1 and D_2 are continuously measured over 250 s as shown in Fig. 10(a). The measured D_1 and D_2 undergo a similar time-dependent trace with a 70 nm drift in peak to valley. The difference between D_1 and D_2 are smaller than the variation of D_1 or D_2 itself by two orders. As shown in

Fig. 10(b), the value of $A \times \Delta(D_2 - D_1)$ is calculated with the coefficient A being 141.41 for correction of the refractive index of air. The corrected relative distance variation $\Delta L/L$ is plotted in Fig. 10(c), where a slowly varying drift of about 70 nm observed in both D_1 and D_2 are well suppressed. The standard deviation of the corrected L is estimated to be 7×10^{-9} , which is lower than the uncertainty limit of the empirical equations. The refractive index of air estimated by $\Delta D_1/L$ is well matched to the change of the refractive index of air Δn calculated by the Ciddor's formula. The difference between two results show a peak-to-valley discrepancy of 2.8×10^{-9} in standard deviation. It implies that the frequency-comb-referenced method performed in this study is able to compensate the refractive index of air with the uncertainty level of 10^{-8} without sensing of ambient air parameters.

6. Summary

In this review, we have dealt with the refractive index of air with the aim of improving the measurement uncertainty of laser-based distance measurements. Fig. 11 summarizes that today's most accurate empirical formulae for prediction of the refractive index of air reaches an uncertainty level of 10^{-8} , being limited by the inaccuracy practically encountered in identifying the ambient air condition. At the same time, the refractometer technology allows for 10^{-9} uncertainty in identifying the true value of the refractive index of air, but this precision can be transferable to actual distance measurement only with extreme environment control. In the meanwhile, the two-wavelength method is an alternative solution to practical compensation of the refractive index of air, its practical application is restricted by the limited choice of the laser wavelengths available today. Nonetheless, recent attempts made by incorporating femtosecond lasers demonstrated the potential of achieving 10^{-9} uncertainty by making the most of the laser frequency stability achieved by locking to the radio-frequency atomic clock. Further improvement is expected in the achievable uncertainty in the two-wavelength interferometry in the near future by generating multiple optical frequencies concurrently from the frequency comb stabilized to optical clocks.

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